UGB 1st Mock for Topper: Solutions

Indian Statistical Institute • B.Stat/B.Math Entrance

Problem 1

Let $d = \gcd(a, b)$ and write $a = dx$, $b = dy$ with $\gcd(x, y) = 1$. Then $ab = d^2xy$, so xy is a perfect square; since x, y are coprime, each of them must itself be a perfect square, say $x = m^2$ and $y = n^2$. Now $a + b = d(m^2 + n^2)$ is prime and d divides it, which forces $d = 1$. Hence $p = a + b = m^2 + n^2$. Since $a \neq b$ we have $m \neq n$ and both are positive, so $p \geq 1^2 + 2^2 = 5$, making p odd. For $m^2 + n^2$ to be odd, m and n must have opposite parity; without loss of generality take $m = 2u$ and $n = 2v + 1$. Then

$$m^2 + n^2 = 4u^2 + 4v^2 + 4v + 1 = 4(u^2 + v^2 + v) + 1,$$

which is of the form $4k + 1$ with $k = u^2 + v^2 + v$. ■

Problem 2

Let $S = \sum_{i=1}^{n} a_i$, so $f(1) = a_0 + S$. The coefficient b_{n+1} in $(f(x))^2$ collects all products a_ia_j with $i + j = n + 1$; since $\deg f = n$, both indices lie in $\{1, \ldots, n\}$, giving

$$b_{n+1} = \sum_{i=1}^{n} a_i\, a_{n+1-i}.$$

The condition $a_k \leq a_0$ gives $a_{n+1-i} \leq a_0$, and so

$$b_{n+1} \leq a_0 \sum_{i=1}^{n} a_i = a_0S.$$

On the other hand, expanding $(f(1))^2 = (a_0 + S)^2 = a_0^2 + 2a_0S + S^2$ and using $a_0, S \geq 0$,

$$\tfrac{1}{2}(f(1))^2 = \tfrac{1}{2}a_0^2 + a_0S + \tfrac{1}{2}S^2 \geq a_0S \geq b_{n+1},$$

which is the required inequality. ■

Problem 3

Label $AB = c$, $BC = a$, so the hypotenuse is $AC = \sqrt{a^2 + c^2}$ and the area is $\frac{1}{2}ac$. The angle bisector from B meets AC at P, and by the Angle Bisector Theorem $\frac{AP}{PC} = \frac{c}{a}$, hence

$$AP = \frac{c}{a + c}\sqrt{a^2 + c^2}.$$

Imposing $AP^2 = [\triangle ABC]$ gives

$$\frac{c^2(a^2+c^2)}{(a+c)^2} = \frac{ac}{2}.$$

Dividing by $c > 0$ and cross-multiplying, $2c(a^2+c^2) = a(a+c)^2$. Expanding both sides and cancelling the common $2a^2c$,

$$2c^3 = a^3 + ac^2 \implies a^3 + ac^2 - 2c^3 = 0.$$

Noting that $a = c$ is a root, we factorise as

$$(a-c)(a^2+ac+2c^2) = 0.$$

The quadratic factor is strictly positive for $a, c > 0$, so $a = c$, and $\triangle ABC$ is an isosceles right triangle. $\blacksquare$

Problem 4

Since $\left(f(x) - \frac{1}{2}\sin x\right)^2 \geq 0$, integrating on $[0, \pi]$ gives

$$\int_0^\pi f(x)^2\,dx - \int_0^\pi f(x)\sin x\,dx + \tfrac{1}{4}\int_0^\pi \sin^2 x\,dx \geq 0.$$

Using $\sin^2 x = \frac{1-\cos 2x}{2}$,

$$\int_0^\pi \sin^2 x\,dx = \left[\tfrac{x}{2} - \tfrac{\sin 2x}{4}\right]_0^\pi = \tfrac{\pi}{2}.$$

Substituting and rearranging,

$$\int_0^\pi f(x)\sin x\,dx \leq \int_0^\pi f(x)^2\,dx + \tfrac{1}{4}\cdot\tfrac{\pi}{2} = \int_0^\pi f(x)^2\,dx + \tfrac{\pi}{8}. \blacksquare$$

Problem 5

By Kummer's theorem, the exponent of a prime p in $\binom{A+B}{A}$ equals the number of carries when adding A and B in base p. Apply this with $p = 2$ and $A = B = n$. Adding $n + n$ in binary is a left shift of n, so a carry is produced at every bit position where n has a 1; that is, the number of carries equals the number of 1s in the binary expansion of n. A positive integer is a power of 2 if and only if its binary expansion contains exactly one 1. Since $n > 1$ and n is not a power of 2, its binary expansion carries at least two 1s, so there are at least two carries. By Kummer's theorem, $2^2 = 4$ divides $\binom{2n}{n}$. $\blacksquare$

Problem 6

From $\frac{p}{q} < \frac{a}{b} < \frac{r}{s}$, clearing denominators gives $aq - bp > 0$ and $rb - as > 0$. All variables are positive integers, so both quantities are integers at least 1: $aq - bp \geq 1$ and $rb - as \geq 1$. Multiplying the given identity $qr - ps = 1$ by b,

$$qrb - bps = b,$$

and adding and subtracting aqs on the left,

$$b = (qrb - aqs) + (aqs - bps) = q(rb - as) + s(aq - bp).$$

Applying the two lower bounds, $b \geq q \cdot 1 + s \cdot 1 = q + s$. $\blacksquare$

Problem 7

Let $P(x) = (x - r_1)(x - r_2)$ with integer roots r_1, r_2, and let $n \in \mathbb{Z}$ satisfy $P(n) = 13$. Then

$$(n - r_1)(n - r_2) = 13,$$

and since 13 is prime, the integer factor pairs are only $(1, 13)$ and $(-1, -13)$.

If $\{n - r_1, n - r_2\} = \{1, 13\}$, say $n - r_1 = 1$ and $n - r_2 = 13$, then

$$P(n + 1) = ((n + 1) - r_1)((n + 1) - r_2) = (2)(14) = 28.$$

If instead $\{n - r_1, n - r_2\} = \{-1, -13\}$, say $n - r_1 = -1$ and $n - r_2 = -13$, then

$$P(n - 1) = ((n - 1) - r_1)((n - 1) - r_2) = (-2)(-14) = 28.$$

In either case, one of $P(n + 1)$ or $P(n - 1)$ is exactly 28. ■

Problem 8

Since f is continuous with $f(0) = 0$ and $f(1) = 1$, the Intermediate Value Theorem guarantees that for each $k \in \{0, 1, \ldots, n\}$ the function attains the value k/n. Define recursively $x_0 = 0$ and, for $k = 1, \ldots, n$,

$$x_k = \min \left\{ x \in [x_{k-1}, 1] : f(x) = \tfrac{k}{n} \right\}.$$

This gives $0 = x_0 < x_1 < \cdots < x_n = 1$ with $f(x_k) = k/n$ for each k.

Applying the Mean Value Theorem on each subinterval $[x_{i-1}, x_i]$, there exists $c_i \in (x_{i-1}, x_i)$ with

$$f'(c_i) = \frac{f(x_i) - f(x_{i-1})}{x_i - x_{i-1}} = \frac{1/n}{x_i - x_{i-1}},$$

so $\dfrac{1}{f'(c_i)} = n(x_i - x_{i-1})$. Summing and telescoping,

$$\sum_{i=1}^{n} \frac{1}{f'(c_i)} = n \sum_{i=1}^{n} (x_i - x_{i-1}) = n(x_n - x_0) = n.$$

The points c_i lie in pairwise disjoint open intervals and are therefore distinct. ■